

World-Model-Based Reinforcement Learning for Robotic Manipulation

A Technical Study of WORLD4RL and WoVR,
and a From-Scratch Reproduction of Their Reliability Mechanisms

Jiaqi Tang

University of Illinois Urbana-Champaign
github.com/TanhJK728/world-model-rl

July 21, 2026

Abstract

Learned world models promise to replace costly and unsafe real-robot interaction with policy optimization performed entirely “in imagination.” The central research question has shifted from *whether* a generative model can imitate real dynamics to *how* reinforcement learning (RL) can stay reliable when those dynamics are imperfect — under hallucination, long-horizon error accumulation, and policy-induced distribution shift. This report first gives a detailed technical account of two works from Chao Yu’s group along this trajectory: WORLD4RL, which establishes diffusion world models as high-fidelity simulators for refining pre-trained manipulation policies, and WoVR, which treats reliability as the first-class problem and regulates the RL-model interface for post-training Vision–Language–Action (VLA) policies. We then present a compact, from-scratch reproduction (continuous-control PPO, a bootstrapped dynamics ensemble, and uncertainty-aware imagined refinement) that isolates the same mechanisms on a controllable **Pendulum-v1** toy and independently rediscovers two of their central premises: reliable imagined refinement depends on the world model’s *coverage* of the policy’s state distribution, and uncertainty signals must be *recalibrated* as the policy drifts.

Contents

1	Motivation and the research question	1
2	World4RL: Diffusion World Models for Policy Refinement	2
2.1	Idea	2
2.2	Method	2
2.3	Experiments	3
2.4	Conclusions	3
3	WoVR: World Models as Reliable Simulators for Post-Training VLA Policies	3
3.1	Idea	4
3.2	Method	4
3.3	Experiments	4
3.4	Conclusions and limitations	5

4 This project: Reliable RL under Imperfect Learned Dynamics	5
4.1 Positioning	5
4.2 Phase 1 — Continuous-control PPO from scratch	5
4.3 Phase 2 — Learned dynamics ensemble (the “world model”)	6
4.4 Phase 3 — Policy refinement inside the world model	7
4.5 Phase 4 — The exploitation gap	8
4.6 Key finding: two regimes, and why the safeguards did not transfer	8
4.7 Limitations and future work	9
Note on scope	10

1 Motivation and the research question

Robotic manipulation policies are typically bootstrapped by imitation learning (IL), whose ceiling is set by the scarcity and narrow coverage of expert data. RL can refine such policies, but on-policy RL needs massive interaction: real-robot training is costly and unsafe, and physics simulators incur a sim-to-real gap. Generative world models offer a third path — a *learned* simulator in which a policy can be improved without touching the real world.

The difficulty is that a learned world model is not a faithful simulator. In closed-loop, autoregressive rollouts, small one-step errors compound with the horizon, and as the policy improves its state-action distribution drifts away from the model’s training data. If RL naively trusts these rollouts, it is incentivized to *exploit systematic model errors* rather than make real task progress: a policy can look excellent in imagination and fail in reality. Both papers below take this as the core problem; our reproduction measures it directly.

2 World4RL: Diffusion World Models for Policy Refinement

Jiang et al., “World4RL: Diffusion World Models for Policy Refinement with Reinforcement Learning for Robotic Manipulation” (2025).

2.1 Idea

World4RL is a two-stage framework: (i) *pre-train* a diffusion world model on multi-task data to capture diverse dynamics, then (ii) *freeze* it and refine an IL-initialized policy entirely inside it with PPO, avoiding online real-world interaction. Unlike prior work that uses generative models for test-time *planning* (e.g. IRASim, NWM) or RSSM-based latent models that produce blurry, drifting rollouts (DiWA), World4RL uses a sharp diffusion backbone to support end-to-end policy optimization.

2.2 Method

The framework has three learned components: a **diffusion transition model** D_θ (dynamics), a **reward classifier** C_ψ (sparse binary success signal), and the **policy** π_ξ (refined with PPO).

Stage 1 — pre-training.

- *Policy*: a Gaussian policy $\pi_\xi(a_t | x_t) = \mathcal{N}(\mu_\xi(x_t), \Sigma_\xi(x_t))$ trained by behavior cloning, $\mathcal{L}_{\text{BC}}(\xi) = -\mathbb{E}_{(x_t, a_t) \sim \mathcal{D}_{\text{exp}}}[\log \pi_\xi(a_t | x_t)]$.
- *Reward classifier*: $r := C_\psi(x_{t+1})$, a ResNet-18 trained with binary cross-entropy on both expert data and pre-trained-policy rollouts, so it recognizes success states the learned policy actually reaches.
- *Diffusion transition model*: following EDM, a preconditioned denoiser $D_\theta(x_\tau; \tau, c) = c_{\text{skip}}^\tau x_\tau + c_{\text{out}}^\tau F_\theta(c_{\text{in}}^\tau x_\tau; c_{\text{noise}}^\tau, c)$ predicts the next observation from a short history $x_{t-T:t}$ and actions. A key design is the **two-hot action encoding** (from DreamerV3): each continuous action dimension a_i is mapped onto its two nearest bins $b_k \leq a_i \leq b_{k+1}$ with weights $t_i[k] = \frac{b_{k+1}-a_i}{b_{k+1}-b_k}$, $t_i[k+1] = \frac{a_i-b_k}{b_{k+1}-b_k}$, a lossless, differentiable representation ($K \approx 21$ bins) that avoids the reconstruction errors of tokenization or latent compression. Crucially, D_θ is trained on a *mixture*: expert demonstrations $\mathcal{D}_{\text{exp}}$, pre-trained-policy rollouts $\mathcal{D}_{\text{rollout}}$, and *uniformly random* rollouts $\mathcal{D}_{\text{rand}}$, so the model sees a broad range of transitions and does not overfit the narrow expert set.

Stage 2 — policy optimization. The frozen D_θ generates imagined rollouts; C_ψ scores each imagined next state as a sparse binary reward; π_ξ is updated with the clipped PPO objective

$$\mathcal{L}_P(\xi) = \mathbb{E}_t[\min(\rho_t(\xi)A_t, \text{clip}(\rho_t(\xi), 1 - \epsilon, 1 + \epsilon)A_t)], \quad \rho_t(\xi) = \frac{\pi_\xi(a_t | x_t)}{\pi_{\xi_{\text{old}}}(a_t | x_t)}, \quad (1)$$

with a squared-error value loss. Because optimizing inside the model tends to push the policy toward out-of-distribution (OOD) actions where the model is unreliable, World4RL adds **controlled exploration**: the policy standard deviation is clipped to a conservative bound $\sigma \leq e^0$ (versus the common $\sigma \leq e^2$), keeping sampled actions near the support of the model’s training data.

2.3 Experiments

World-model fidelity (Meta-World video prediction). With 330M parameters, World4RL attains the best FVD/FID/LPIPS among NWM, iVideoGPT, and DiWA under both policy and random rollouts (Table 1), and faithfully models even *failure* trajectories where baselines hallucinate success.

Table 1: World4RL video-prediction fidelity on Meta-World (lower is better).

Model	FVD		FID		LPIPS	
	Policy	Random	Policy	Random	Policy	Random
World4RL	326.5	400.1	17.1	23.4	0.019	0.025
NWM	547.4	851.9	30.5	34.9	0.027	0.026
iVideoGPT	450.3	531.3	18.7	20.7	0.026	0.028
DiWA	803.6	1231.0	62.9	96.5	0.080	0.136

Policy learning (Meta-World, sparse reward, 6 tasks, 3 seeds). World4RL reaches the best average success rate 67.5%, a +16-point absolute gain over the BC base (51.5%), beating IL (BC, Diffusion Policy), offline RL (TD3+BC, IQL), and world-model baselines (IRASim, DiWA,

TD-MPC2). It matches offline-to-online methods (RLPD, Uni-O4) while using *no* online samples, versus their 346k/470k online steps.

Real robot (Franka Panda, 6 tasks, 20 trials each). World4RL reaches 93.3% average success, a +25-point gain over BC (68.3%) and above Diffusion Policy (88.3%), and executes tasks more decisively.

Ablations. (i) Two-hot encoding beats linear/one-hot/VQ-VAE/tokenization on all fidelity metrics. (ii) Removing *action-std clipping* or *random rollouts* both degrade policy performance and increase variance — validating that **controlled exploration** and **broad action-space coverage** are critical to stable imagined refinement.

2.4 Conclusions

Diffusion world models can serve as high-fidelity, real-time simulators enabling consistent PPO refinement of pre-trained policies under sparse rewards, without online interaction. The authors explicitly flag the open direction of “more robust RL algorithms that can effectively learn under imperfect learned dynamics.”

3 WoVR: World Models as Reliable Simulators for Post-Training VLA Policies

Jiang, Zhou et al., “WoVR: World Models as Reliable Simulators for Post-Training VLA Policies with RL” (2026).

3.1 Idea

WoVR reframes world-model-based RL as a *reliability* problem rather than a modeling problem. It defines **hallucination** as a systematic mismatch between imagined and real outcomes in closed-loop interaction, driven by two compounding effects: *autoregressive feedback* (the model conditions on its own generated frames, amplifying early errors) and *distribution shift* (the evolving policy drifts from the world model’s training data). WoVR controls hallucination at three interconnected levels rather than assuming a faithful simulator.

3.2 Method

(1) Stabilized action-conditioned world model. Built on the Wan2.2-TI2V-5B video-diffusion backbone and reformulated as an action-conditioned generator. Actions are injected through a **dual-channel** design (timestep-conditioned normalization plus replacing text embeddings in cross-attention), giving frame-level control while preserving the DiT structure. A **first-frame-anchored** context $[o_0, o_{t-c:t}]$ fixes global layout while memory frames preserve local dynamics, and **noisy context augmentation** corrupts non-reference context frames during training so the model is robust to its own generated frames at inference. It is trained with a rectified-flow objective $\mathcal{L} = \mathbb{E} \|u(x_\tau, c, \tau; \phi) - (x_1 - x_0)\|^2$ with $x_\tau = (1 - \tau)x_0 + \tau x_1$. Reward is supplied by either a lightweight ResNet (sparse binary) or a Qwen3-VL model (dense ordinal progress $\ell_t \in \{0, \dots, 10\}$).

(2) Keyframe-Initialized Rollouts (KIR). Instead of always starting imagined rollouts from the episode start o_0 — which forces the model to imagine a long, drift-prone prefix before reaching decisive contacts — KIR initializes a portion of rollouts from *keyframes* o_k near task-critical (often failure) states, using $[o_0, o_{T_{\text{KIR}}-3:T_{\text{KIR}}}]$ as context. This *shortens the effective prediction depth*, reducing hallucination compounding. The policy is updated with GRPO,

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{T_i^{\text{valid}}} \sum_{t=1}^{T_i^{\text{valid}}} \min(\rho_t^{(i)}(\theta) \hat{A}^{(i)}, \text{clip}(\rho_t^{(i)}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}^{(i)}) \right], \quad (2)$$

where $\hat{A}^{(i)}$ is a group-relative advantage and T_i^{valid} counts steps up to first success; trajectory-length normalization lets short, task-critical (KIR) segments dominate the gradient over long drift-prone continuations.

(3) PACE: policy-aligned co-evolution. Because policy optimization shifts the action distribution away from the world model’s training data, WoVR periodically re-aligns them: train an initial WM_{Base} on base-policy trajectories, optimize the policy inside it, then collect a *small* set of rollouts under the evolved policy to refine the model into WM_{Evo} . This is *low-frequency* (unlike classical MBRL’s high-frequency model updates), so it needs no continuous supervision while restoring simulator reliability.

3.3 Experiments

World-model quality (LIBERO, 512-frame closed-loop rollouts). WoVR outperforms EVAC, Cosmos-Predict2, and OpenSora on LPIPS/FID/FVD/FloLPIPS, with the margin *growing* at longer horizons, at 23 FPS (vs OpenSora’s 7) despite a larger backbone — via only five diffusion steps and a 3D VAE.

Policy performance (LIBERO success rate). Table 2 summarizes the headline result under two SFT initializations. WoVR beats limited-budget online GRPO and the WMPO imagined-RL baseline in every suite, with the largest gains where the base policy is weakest.

Table 2: WoVR LIBERO success rates (%). All methods share a 2,500-trajectory budget.

Method	Spatial	Object	Goal	Long	Avg
<i>One-trajectory SFT</i>					
OpenVLA-OFT (base)	63.6	36.4	48.2	13.8	40.5
+ GRPO (online)	66.6	45.2	52.2	14.6	44.6
+ WMPO	67.8	65.4	56.6	13.8	50.9
+ WoVR	84.2	80.8	77.4	35.8	69.5
<i>Full-trajectory SFT</i>					
OpenVLA-OFT (base)	93.6	83.0	90.0	85.6	88.1
+ WMPO	95.0	94.8	92.8	87.0	92.4
+ WoVR	98.8	98.8	94.8	91.4	96.0

Real robots. On a Franka Panda, WoVR improves the base policy by +28.9 points (vs WMPO’s +13.4); on the noisier AgileX Piper by +13.4 (vs WMPO’s +3.4), showing that naive imagined RL remains vulnerable to hallucination while WoVR transfers reliably. The framework also generalizes to a $\pi_{0.5}$ backbone (+22.3 points).

3.4 Conclusions and limitations

Hallucination in closed-loop imagined interaction is the central reliability bottleneck of world-model-based RL for VLA; controlling it at the simulator, interaction, and alignment levels lets a learned model act as a practical RL simulator. The authors note their analysis is empirical — no formal characterization of how hallucination propagates into policy optimization, and no regret bound of the imagined-optimal policy relative to the real-optimal one — and that longer-horizon and mobile manipulation remain open.

4 This project: Reliable RL under Imperfect Learned Dynamics

4.1 Positioning

World4RL and WoVR operate at the frontier: diffusion/video world models, VLA policies, real robots. To *understand the mechanisms* rather than the engineering, this project builds the entire stack from scratch in PyTorch on the controllable **Pendulum-v1** state space, where the exploitation gap can be measured exactly. It is a state-space, dynamics-model MBRL study (in the lineage of PETS, MBPO, and Dreamer’s imagination step), not a generative video model; the transferable content is the *reliability mechanism*. The pipeline has four phases.

4.2 Phase 1 — Continuous-control PPO from scratch

Pendulum-v1 has observation $s_t = (\cos \theta_t, \sin \theta_t, \dot{\theta}_t)$, torque $a_t \in [-2, 2]$, and reward $r_t = -(\theta_t^2 + 0.1 \dot{\theta}_t^2 + 0.001 a_t^2)$. The policy is a tanh-squashed Gaussian: sample $z \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ and set $a = a_{\text{scale}} \tanh(z) + a_{\text{bias}}$, with the change-of-variables correction

$$\log \pi_\theta(a \mid s) = \log \mathcal{N}(z; \mu_\theta, \sigma_\theta^2) - \sum_i \log(a_{\text{scale}}(1 - \tanh^2 z_i)). \quad (3)$$

Advantages use GAE- λ : with $\delta_t = r_t + \gamma V(s_{t+1})(1 - d_{t+1}) - V(s_t)$, $\hat{A}_t = \sum_{l \geq 0} (\gamma \lambda)^l \delta_{t+l}$, and the clipped PPO objective is optimized with a clipped value loss and target-KL early stopping.

A reliability lesson already in Phase 1. Out of the box the from-scratch PPO plateaus at ~ -1000 (barely above random). The cause is scale: unnormalized Pendulum returns are $O(10^3)$, so the critic loss is ~ 2000 – 4000 , giving noisy advantages. Adding **reward normalization** to the training signal only (observations untouched, so the downstream world-model pipeline is unaffected) drops the value loss to ~ 0.05 and solves the task. Table 3 reports the final policy.

4.3 Phase 2 — Learned dynamics ensemble (the “world model”)

The world model is an ensemble of $K = 5$ MLPs predicting the *state delta*,

$$\hat{s}_{t+1}^{(j)} = s_t + f_{\theta_j}(s_t, a_t), \quad \Delta s_t = s_{t+1} - s_t, \quad j = 1, \dots, K, \quad (4)$$

Table 3: Phase 1 deterministic evaluation (30 episodes). Config: 10^6 steps, constant LR, entropy coef. 0.01, reward normalization.

Policy	Return (mean $\pm$ std)
PPO seed 0	-148.7 ± 94.4
PPO seed 1	-152.7 ± 100.9
PPO seed 2	-157.5 ± 100.5
Across seeds	-153.0 ± 4.4
Random baseline	-1198.6 ± 280.7

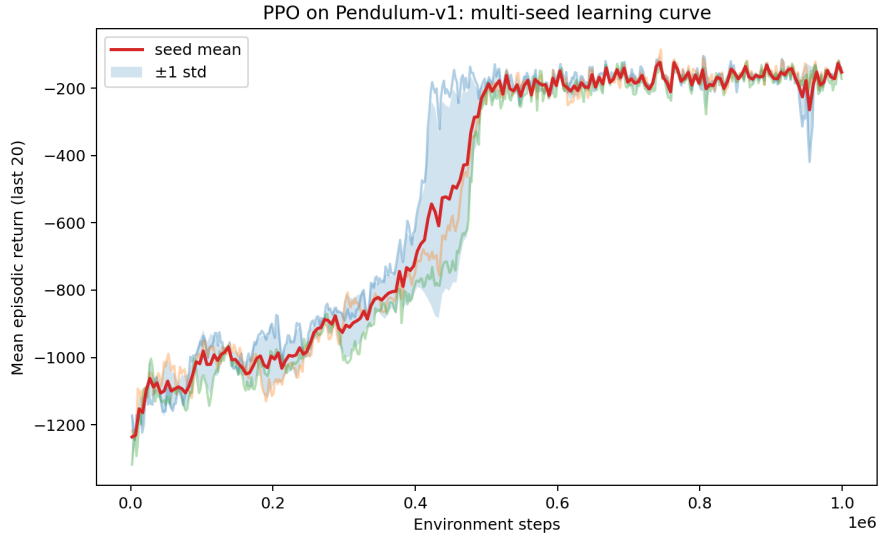


Figure 1: Multi-seed PPO learning curve: a sharp phase transition near 400–500k steps, then a stable ~ -150 plateau.

with input/target normalization and a re-projection of $(\cos \theta, \sin \theta)$ onto the unit circle after each step. Three deliberate choices mirror the papers’ concerns: (i) *no learned reward* — imagined rollouts use the exact Pendulum reward, isolating exploitation to dynamics error; (ii) *bootstrap ensemble + episode-level train/val split*; (iii) *ensemble disagreement as epistemic uncertainty*,

$$u_t = \text{mean}_k \text{Var}_j [\hat{s}_{t+1,k}^{(j)}], \quad (5)$$

calibrated into validation quantiles ($q_{95} = 6.63 \times 10^{-6}$, the default rollout-termination threshold).

Result: accurate on-distribution, unreliable off-distribution. Trained only on *expert-policy* data, the model’s error explodes under distribution shift (Table 4), and ensemble uncertainty rises in the same order — a usable trust signal. This is the toy-scale analogue of the hallucination that World4RL’s random rollouts and WoVR’s stabilized model are designed to suppress.

4.4 Phase 3 — Policy refinement inside the world model

A vectorized imagined environment applies the exact reward, uses the ensemble mean as the next state, and the ensemble variance as u_t . A real PPO checkpoint warm-starts the policy (RL post-training in imagination). Four methods are compared:

Table 4: Phase 2 world-model error by data source (model trained on **policy** only).

Source	one-step MSE	vs policy	mean uncertainty
policy	1.96×10^{-6}	$1\times$	1.2×10^{-6}
random	2.97×10^{-3}	$\sim 1500\times$	2.2×10^{-4}
ood	5.17×10^{-3}	$\sim 2600\times$	3.9×10^{-4}

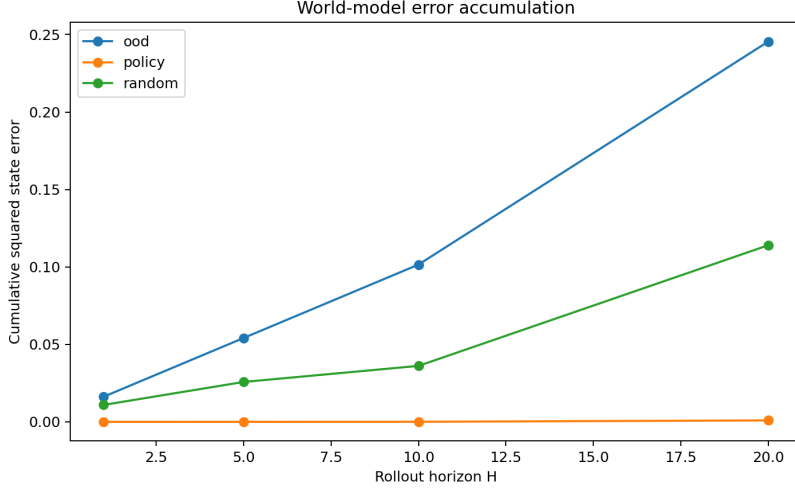


Figure 2: Cumulative rollout error stays flat on-distribution and diverges off-distribution, compounding with horizon — the toy analogue of long-horizon hallucination.

- **Fixed $H=5$ / $H=20$:** fixed imagined-rollout length (longer H = stronger credit assignment but more error accumulation).
- **Uncertainty termination:** cut the rollout when $u_t > \tau$ ($\tau = q_{95}$), an *adaptive* horizon $H_t = \min\{H_{\max}, \text{first } t : u_t > \tau\}$ — the toy counterpart of WoVR’s KIR error-depth control.
- **Weighted advantage:** down-weight uncertain transitions, $\tilde{A}_t = w_t \hat{A}_t$, $w_t = \max(0.02, \exp(-\beta u_t / \tau))$ — confidence-weighted updates.

4.5 Phase 4 — The exploitation gap

Because short imagined-segment returns are not comparable across horizons, every policy is evaluated from *matched* initial states for a common 200-step horizon in both worlds:

$$J_{\text{model}}(\pi) = \frac{1}{N} \sum_i R_{\text{model}}^{(i)}, \quad J_{\text{real}}(\pi) = \frac{1}{N} \sum_i R_{\text{real}}^{(i)}, \quad \textbf{Exploitation Gap} = J_{\text{model}}(\pi) - J_{\text{real}}(\pi). \quad (6)$$

A positive gap means the policy looks better in imagination than in reality — it is exploiting model error. Two regimes are studied.

Regime A — refine the in-distribution (near-optimal) policy. Warm-starting from the same policy that generated the model’s data, the model is so faithful that $\text{model} \approx \text{real}$ everywhere (gap ≈ 0 for $H=20$ and uncertainty termination). There is nothing to exploit and no headroom to improve — a degenerate regime.

Regime B — refine an off-distribution (sub-optimal) policy. Warm-starting instead from a deliberately sub-optimal checkpoint (≈ -570), 500k imagined steps, 3 seeds, against the *same* expert-trained model, the exploitation gap appears (Table 5) and is largest at $H=20$ — consistent with more error accumulation over longer rollouts.

Table 5: Phase 4, Regime B: 200-step model-vs-real comparison (mean $\pm$ std over 3 seeds).

Method	Real return	Exploitation gap	Mean horizon
mid init (baseline)	-576	-3	—
Fixed $H=5$	-792 ± 249	$+20 \pm 22$	5.0
Fixed $H=20$	-754 ± 105	$+28 \pm 21$	20.0
Uncertainty termination	-1473 ± 24	$+9 \pm 26$	4.3
Weighted advantage $H=20$	-1462 ± 4	-13 ± 1	20.0

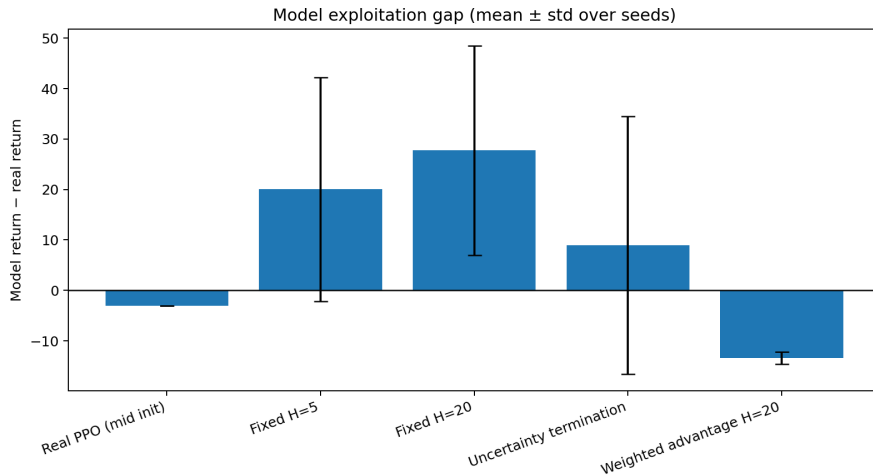


Figure 3: Exploitation gap across seeds (Regime B). The gap is largest for fixed long-horizon rollouts.

4.6 Key finding: two regimes, and why the safeguards did not transfer

Combining the regimes gives a sharper statement than either alone. (1) In-distribution refinement is safe but uninformative. (2) Off-distribution refinement produces a genuine, horizon-growing exploitation gap under fixed rollouts — *and* the uncertainty machinery, calibrated on the narrow expert distribution ($q_{95} = 6.6 \times 10^{-6}$), does not transfer: the sub-optimal policy induces uncertainty $\sim 10^{-5}$ almost everywhere, so termination fires immediately (adaptive horizon collapses $20 \rightarrow 4.3$), starving credit assignment, while weighted advantages down-weight nearly every transition. Both “safeguards” therefore *hurt* real return here.

The lesson is not “these methods fail,” but the more precise and useful one: **a static world model supports reliable policy improvement only within its training distribution, and uncertainty thresholds must be calibrated on the distribution the policy actually visits.** This is exactly long-horizon error accumulation and policy-induced distribution shift made measurable on a controllable toy — and it independently points at the two papers’ remedies.

Table 6: Mechanism-level correspondence between this reproduction and the two papers.

Reproduction finding	World4RL	WoVR
Off-distribution refinement collapses; model needs coverage of visited states	Train the world model on expert $\cup$ policy $\cup$ <i>random</i> rollouts; broader action-space coverage	Distribution shift as a core failure mode
Fixed long horizon $\Rightarrow$ larger exploitation gap	—	KIR shortens effective error depth
Narrow-distribution uncertainty thresholds stop transferring after policy shift	Controlled exploration (std clipping) keeps the policy in support	PACE co-evolves the model with the evolving policy

4.7 Limitations and future work

The decisive next step is to train the ensemble on *broad* replay data (e.g. policy $\cup$ random) and recalibrate uncertainty there — the toy analogue of World4RL’s random rollouts and WoVR’s PACE — which is expected to let the uncertainty-aware methods recover real return and shrink the gap. Iterative data collection (MBPO-style) is the principled cure for policy-induced shift; Regime A should also be seeded before drawing quantitative conclusions. Everything here is a 3-D state space, but the mechanisms (delta dynamics, ensemble uncertainty, imagined refinement, exploitation-gap measurement) transfer to higher-dimensional and pixel/latent world models.

Note on scope

This is a focused, from-scratch reproduction and mechanism study built to understand how policy optimization stays reliable under an imperfect learned model. It reproduces state-space dynamics-model MBRL, not a generative or video world model, and reports results — including a safeguard that fails to transfer off-distribution — as measured. Code and figures: <https://github.com/TanhJK728/world-model-rl>.

References

- [1] Z. Jiang, K. Liu, Y. Qin, S. Tian, Y. Zheng, M. Zhou, C. Yu, H. Li, D. Zhao. *World4RL: Diffusion World Models for Policy Refinement with Reinforcement Learning for Robotic Manipulation*. arXiv:2509.19080, 2025.
- [2] Z. Jiang, S. Zhou, et al. *WoVR: World Models as Reliable Simulators for Post-Training VLA Policies with RL*. arXiv:2602.13977, 2026.
- [3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov. *Proximal Policy Optimization Algorithms*. arXiv:1707.06347, 2017.
- [4] J. Schulman, P. Moritz, S. Levine, M. Jordan, P. Abbeel. *High-Dimensional Continuous Control Using Generalized Advantage Estimation*. ICLR, 2016.

- [5] K. Chua, R. Calandra, R. McAllister, S. Levine. *Deep Reinforcement Learning in a Handful of Trials using Probabilistic Dynamics Models (PETS)*. NeurIPS, 2018.
- [6] M. Janner, J. Fu, M. Zhang, S. Levine. *When to Trust Your Model: Model-Based Policy Optimization (MBPO)*. NeurIPS, 2019.
- [7] D. Hafner, J. Pasukonis, J. Ba, T. Lillicrap. *Mastering Diverse Control Tasks through World Models (DreamerV3)*. Nature, 2025.
- [8] Z. Shao et al. *DeepSeekMath: Pushing the Limits of Mathematical Reasoning (GRPO)*. arXiv:2402.03300, 2024.
- [9] T. Karras, M. Aittala, S. Laine, S. Aila. *Elucidating the Design Space of Diffusion-Based Generative Models (EDM)*. NeurIPS, 2022.